

AIRA JOSH C. YNTE

CURRICULUM VITAE

Software & AI Engineer · Computer Engineering Graduate

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Muntinlupa, Metro Manila, Philippines

PROFESSIONAL SUMMARY

Detail-oriented Computer Engineering graduate (Expected July 2026) specializing in software architecture, systems design, and AI/IoT integration. Experienced in full-stack web development, digital logic verification, embedded firmware development, and machine learning pipelines. Passionate about designing responsive applications, optimizing signal acquisition systems, and deploying intelligent solutions.

EDUCATION

Bachelor of Science in Computer Engineering

2022 – July 2026 (Expected)

Colegio de Muntinlupa — Muntinlupa City, Metro Manila, Philippines

Specialization: Software Architecture, Systems Design, and Embedded Systems

- Coursework in database systems, computer network architectures, software engineering, digital logic circuits, microprocessors, and foundations of artificial intelligence.
- Academic projects focused on bio-signal processing, real-time telemetry pipelines, and custom PCB and hardware design.

PROFESSIONAL EXPERIENCE

Freelance Software & AI Developer

2023 – Present

Various Clients

- Integrated advanced language models and LLM agent chains for document parsing, achieving 98.2% data extraction accuracy from unstructured datasets.
- Built computer vision tracking pipelines utilizing MediaPipe BlazePose to analyze real-time video streams and execute pose tracking on browser-side applications.
- Developed real-time IoT monitoring dashboards and telemetry visualization tools using Flutter, Firebase, and relational databases.
- Configured model orchestration prompts and Retrieval-Augmented Generation (RAG) pipelines to search and query internal document stores.
- Optimized backend image processing and data loading flows, reducing asset size and latency by 65%.

IC Design Intern (Design & Verification)

2025

Xinyx Design Consultancy and Services Inc.

- Designed and verified AMBA APB3-compliant peripheral interfaces using SystemVerilog RTL, writing testbenches that verified timing constraints and data lines during behavioral simulations.
- Gained hands-on experience in digital logic design, circuit simulation, and semiconductor design/verification workflows.
- Practiced systematic troubleshooting and rigorous logical testing for complex system components.

PROJECTS SHOWCASE

AeroVit — Gamified AI Fitness Platform ([Live Web](#))

September 2024 – April 2026

Role: Lead Developer & Hardware Architect

- Developed a cross-platform mobile application in Flutter (Dart) using Provider state management, Firestore, and local cache.
- Built a 33-landmark MediaPipe BlazePose pipeline with EMA noise-filtering and joint-angle state machines for real-time fitness coaching.
- Designed custom smartwatch PCBs using ESP32-S3 and routed traces for BLE 5.0, MAX30102 (HR/SpO2), and QMI8658 IMU over I2C/SPI.
- Deployed AERO ERC-20 token rewards on Sepolia Testnet with an off-chain ledger sync and 5% token burn on withdrawal.
- Developed a turn-based dungeon crawler (Flame 2D engine) alongside Next.js product showcase and Vite technical guides.

Stack: ESP32-S3, C/C++, Flutter, Dart, Flame, Firebase, MediaPipe, BLE 5.0, Next.js, Ethereum Sepolia, ERC-20

Fehuvia — Open Finance Web3 B2B SME Treasury Workstation [\(Live Web | Repo\)](#)

2026

Role: Lead Web3 & Full-Stack Developer

- Coded and deployed Solidity smart contracts on Morph L2 for atomic, non-custodial settlements (sub-2s clearance using mUSDC).
- Designed real-time Node.js listener daemon syncing smart contract events to PostgreSQL, maintaining financial state consistency.
- Built backend integrating context-aware GPT-4o RAG and ChromaDB, parsing uploaded invoice PDFs with 98.2% OCR accuracy.
- Created modern glassmorphic dashboard in React 19 / Tailwind CSS, utilizing Recharts to visualize cashflow runtimes.
- Implemented client-side signing modules using MetaMask (EIP-1193) and set up self-healing background gas faucet node.

Stack: React 19, Vite, Tailwind CSS, Recharts, Express, PostgreSQL, OpenAI GPT-4o, Ethers.js, Solidity, Hardhat, Morph L2

StickOut — Interactive VLSI EDA Editor [\(Live Web | Repo\)](#)

May 2026 – June 2026

Role: Creator & Lead Frontend Engineer

- Developed interactive EDA viewport using raw HTML5 Canvas Context 2D, implementing screen-to-world coordinate transforms, grid snapping, and dynamic zooming (0.1x to 5.0x).
- Programmed auto-bridging algorithm for same-layer orthogonal wire intersections, supporting right-click overrides for solid junctions.
- Built Photoshop-style layers panel supporting visibility toggles, lock status, and opacity mixing.
- Created custom double-pass LaTeX label parser to render math subscripts (e.g. V_{DD} , V_{SS}) using regex and text metrics.
- Coded high-resolution image exporting at 2x scale mapping layout bounding box, and managed state with 50-level Undo/Redo stack.

Stack: React 19, Vite 8, HTML5 Canvas, Pure CSS3, Lucide Icons, LaTeX Rendering, VLSI Layout

The Safehouse — Community Showcase & Pitch Site [\(Live Web | Repo\)](#)

2025 – Present

Role: Frontend & 3D Web Developer

- Built interactive 3D journey scene in React Three Fiber with procedural terrain rendering and camera motion tracking.
- Wired GSAP timelines and ScrollTrigger to Lenis smooth scroll for 13 narrative scroll sections, synced via shared Tempus clock.
- Implemented accessibility-first MotionModeContext with static HTML/CSS fallbacks to respect user motion preferences.
- Integrated Rive vector animations and Framer Motion primitives to deliver smooth component-level transitions.

Stack: React 19, Vite, Tailwind CSS 4, Three.js, @react-three/fiber, GSAP, Framer Motion, Lenis, Rive

Plant.io — 8-Bit IoT Nursery Dashboard

2025

Role: Full-Stack Mobile & IoT Developer · Freelance

- Developed a cross-platform Flutter dashboard utilizing Provider state management to display real-time sensor streams (moisture, temperature, pH, EC).
- Configured low-latency commands over Firebase Realtime DB to actuate water pumps and fans in under 150ms.
- Created offline-first telemetry caching layer using SharedPreferences, implementing local push alerts.
- Integrated Open-Meteo weather API to fetch location-specific agricultural recommendations, and styled retro 8-bit theme (Press Start 2P fonts, CRT scanlines, retro SFX) with an animated plant avatar.

Stack: Flutter, Dart, Provider, Firebase Realtime DB, Cloud Firestore, fl_chart, Open-Meteo API

EMG Interface Controller [\(Repo\)](#)

2025

Role: Embedded Developer & Hardware Designer · Academic Project

- Programmed ESP32-S3 microcontroller to sample analog EMG signals at 500 Hz and executed multi-stage DSP noise-reduction pipeline.
- Coded user-specific adaptive threshold calibration logic to normalize signal envelopes.
- Developed scientific visualization GUI in Python (Matplotlib and NumPy) to read and plot serial data streams in real time.
- Built game controller bridge (Roblox integration via pynput) translating muscle contractions into keyboard inputs.
- Designed custom PCB in KiCad and 3D-printed enclosure in Fusion 360 to package the controller.

Stack: ESP32-S3, C/C++, Python, KiCad, Fusion 360, NumPy, Matplotlib, pynput

Student Consultation System

2025

Role: Full-Stack Developer · Freelance

- Developed role-based dashboards in React/Vite supporting distinct scheduling channels for students, professors, and admins.
- Secured system endpoints with JWT authentication, bcrypt password hashing, and parameterized SQL queries to prevent SQL injections.
- Integrated Sharp image-processing pipeline to crop and compress profile photos, reducing asset size by 65%.
- Implemented real-time in-app messaging capabilities enabling student-professor chats with file attachments.

Stack: React, TypeScript, Vite, Tailwind CSS, Node.js, Express, PostgreSQL, JWT, Multer, Sharp

Walang Basagan ng Thrift — E-Commerce Platform

2025

Role: Full-Stack Web Developer · Freelance

- Designed storefront with Y2K aesthetic, pink-and-white rounded UI, background music, custom catalogs, shopping carts, and order history pages.
- Built Express.js server supporting SQLite (better-sqlite3) and JSON file fallbacks for database portability.
- Created admin/moderator back-office dashboard to manage products, users, orders, homepage promotions, and support tickets.
- Programmed customer support widget with Assignment and Threading logic for assigning support tickets to moderators.

Stack: React 19, TypeScript, Vite, Tailwind CSS, Express, SQLite, multer, bcryptjs

Manhwa Reader — Web Application

2026

Role: Frontend Developer · Personal Project

- Built single-page web app browsing manga content via MangaDex REST API, filtering duplicate chapter files.
- Managed inconsistent API payloads with defensive default states and local storage caching.
- Implemented theme context with localStorage persistence for light/dark mode and built in-reader chapter selector.

Stack: React 19, TypeScript, Vite, React Router DOM, Tailwind CSS, MangaDex API

Aira.sys Personal Portfolio [\(Live Web\)](#) [\(Repo\)](#)

2025 – Present

Role: Solo Designer & Developer

- Designed terminal-OS layout using React and Tailwind CSS with a persistent sidebar, scroll-spy navigation, and system headers.
- Built interactive 3D portfolio grids, custom cards with perspective transform animations, and inline SVG system architecture diagrams.
- Deployed as static website to GitHub Pages via automated GitHub Actions pipelines.

Stack: React, TypeScript, Vite, Tailwind CSS, Framer Motion, GitHub Pages

SKILLS & COMPETENCIES

AI & Machine Learning: OpenAI API, MediaPipe BlazePose, Prompt Engineering, Retrieval-Augmented Generation (RAG), Vector DBs (ChromaDB), OpenCV, NumPy, Pandas, Scikit-learn

Programming Languages: TypeScript, JavaScript, Python, Solidity, SQL (PostgreSQL, SQLite, MySQL), C/C++, SystemVerilog, HTML5, CSS3

Web & App Frameworks: React, Next.js, Node.js, Express, Flutter, Dart, Flame 2D, Tailwind CSS, Recharts, Redux, Bootstrap

Tools & Hardware: Git, GitHub, Vite, Docker, GitHub Actions, Hardhat, Ethers.js, MetaMask (EIP-1193), KiCad, Fusion 360, Logic Simulation, REST APIs, JWT, Postman, Linux, Figma, Adobe Suite

Engineering Methodologies: Digital IC Design & Verification, DSP pipelines, Software Architecture, MVC Pattern, Real-Time Telemetry, Zero-Custody Cryptography, Checks-Effects-Interactions Contract Patterns

CERTIFICATIONS & TRAINING

AI Fluency: Framework & Foundations (Anthropic) · Blockchain4Youth (Bitget) · The AI Engineer Path (Scrimba) · Learn React (Scrimba) · Learn TailwindCSS (Scrimba) · Basic Web Development Workshop (Zuitt) · On-the-Job Training Certificate (Xinyx Design Engineering) · Understanding EDR Cybersecurity

ONLINE PRESENCE & LINKS

- **Online Portfolio:** aera0908.github.io
- **GitHub Profile:** github.com/Aera0908
- **LinkedIn Profile:** linkedin.com/in/aira-josh-ynte
- **LeetCode Profile:** leetcode.com/u/Aera0908
- **AeroVit Project:** aerovit.dev
- **Fehuvia Workstation:** fehuvia.app
- **StickOut EDA Editor:** stickout.vercel.app
- **The Safehouse Showcase:** safehouse-inky.vercel.app